

APPENDIX

This appendix presents supplementary numerical results to further evaluate our proposed optimisation model, including an alternative trade-off solution and an analysis of EV charging demand.

A. Revenue-Preserving Trade-off Analysis

The provider-oriented objective maximises charging revenue without explicitly minimising the associated driver costs. Consequently, multiple solutions may achieve the same maximum revenue while producing different driver costs.

To investigate this possibility, an alternative optimisation scenario is considered in which the maximum revenue is maintained while the remaining driver costs are further minimised.

As illustrated in Fig. A.1, the alternative solution achieves the same maximum revenue of AUD 14,745 while further reducing both waiting and driving costs. These results demonstrate that additional improvements in driver outcomes can be achieved without compromising the maximum provider revenue in the investigated scenario.

B. Analysis of EV Charging Demand

To investigate the influence of EV charging demand on the optimisation outcomes, two additional demand scenarios are tested, corresponding to 50% and 150% of the baseline EV demand.

Figs. A.2 and A.3 compare the provider-oriented and driver-oriented optimisation outcomes under the two demand scenarios, respectively. The results indicate that increasing EV charging demand reduces the difference in charging revenue between the provider-oriented and driver-oriented solutions for the investigated scenarios. This observation highlights the trade-off between provider revenue and driver costs to the different level of EV charging demand.

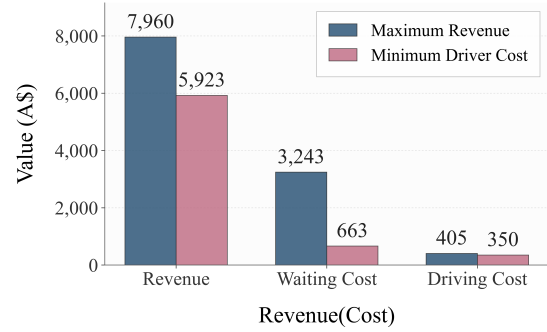


Fig. A.2. Comparison of provider revenue and driver costs under the provider-oriented and driver-oriented objectives with 50% of the baseline EV demand.

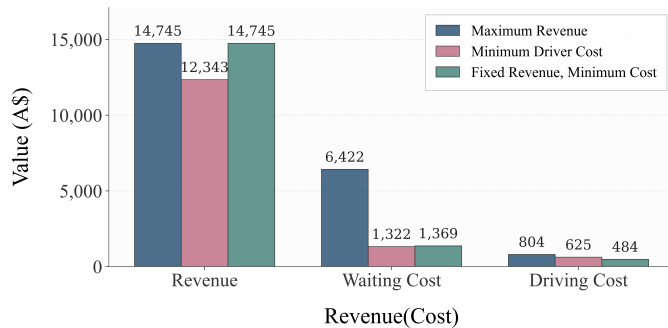


Fig. A.1. Comparison of provider revenue and driver costs under the provider-oriented, driver-oriented, and revenue-preserving trade-off objectives.

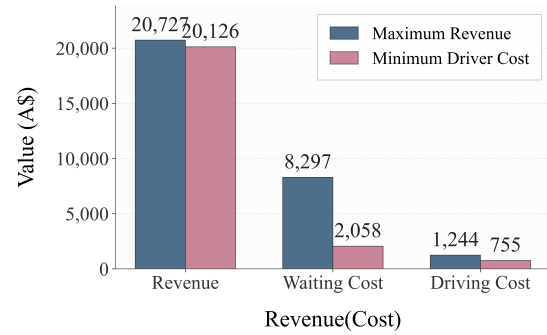


Fig. A.3. Comparison of provider revenue and driver costs under the provider-oriented and driver-oriented objectives with 150% of the baseline EV demand.